MACHINE LEARNING TIME SERIES FORECASTING: A COMPREHENSIVE SURVEY AND STOCK MARKET APPLICATION

by

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(Under the Direction of Khaled Rasheed)

ABSTRACT

This thesis presents two complementary studies that advance the understanding and application of machine learning techniques in time series forecasting, with a particular focus on financial markets. A comprehensive survey identifies top-performing techniques across tree-based models, deep learning architectures, and hybrid approaches. Building on these insights, the thesis applies a specialized forecasting framework to the domain of day trading. By leveraging a combination of LightGBM models with an extensive set of engineered features ranging from multi-timeframe technical indicators to contextual stock attributes, the model uses two years of second-by-second trade and quote data to estimate risk-reward ratios over multiple forward time horizons. Simulated results using realistic execution constraints demonstrate a pronounced performance advantage over human day traders, yielding daily returns several orders of magnitude higher.

INDEX WORDS: TIME SERIES FORECASTING, DAY TRADING, STOCK MARKET, FINANCIAL MARKETS, MACHINE LEARNING, TREE-BASED, LIGHTGBM

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B.S., University of Georgia, 2024

A Thesis Submitted to the Graduate Faculty of the
University of Georgia in Partial Fulfillment of the Requirements for the Degree.

MASTER OF SCIENCE

ATHENS, GEORGIA

2025

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ACKNOWLEDGEMENTS

I would like to thank my major professor, Dr. Rasheed Khaled, for his insight and guidance throughout the writing of this thesis. I am also deeply grateful to my advisory committee, Frederick Maier and Jin Lu, for their support and thoughtful feedback on both coursework and thesis work. I would like to thank all my undergraduate and graduate instructors for equipping me with the knowledge and skills necessary to undertake this research. Lastly, I am forever thankful to my parents for their unwavering support and encouragement throughout my academic journey. This thesis would not have been possible without the mentorship, love, and support I have received along the way.

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CHAPTER 1

Introduction

1.1 Motivation

Time series prediction is a widely recognized and important application of machine learning (ML). Time series prediction usually relies on using historical patterns to accurately forecast future data points. Its significance exists across many domains, including finance, healthcare, energy management, and weather forecasting. In each of these domains, the ability to accurately model and forecast future values holds potential to optimize decision-making, improve performance, or mitigate risks. Among all time series prediction tasks, forecasting stock market behavior is one of the most difficult. This is due to the many complex factors including the countless individuals and institutions acting on the market at the same time. Within this domain, day trading, where positions are opened and closed within the same trading day poses an even greater challenge.

This difficulty, however, also presents an opportunity. While human traders are constrained by cognitive biases, limited processing capacity, and delayed reaction times, ML systems offer the potential to systematically uncover patterns that may be invisible to human traders. In this thesis, I explore the premise that advanced time series prediction ML methods, through their capability to process large amounts of data, identify subtle patterns, and execute rapid decision-making, offer compelling advantages over traditional human expertise in time series prediction, particularly in the domain of financial trading.

1.2 Purpose and Scope

My central aim of this study is to extensively explore state-of-the-art ML methodologies for time series prediction and apply the most promising methods to day trading forecasting in the stock market. This work is structured as a manuscript-style thesis comprising two major studies.

The first paper, *A Survey of Machine Learning Methods for Time Series Prediction*, investigates the landscape of ML techniques applied to time series forecasting. In it, I seek to identify which models offer the most promise across different forecasting formats and tasks. Unlike many existing surveys, this study narrows its focus to research that compares state-of-the-art models within the same experimental setup, allowing for more meaningful insights into their relative strengths, weaknesses, and situational advantages.

The second paper, *Can AI Beat Human Traders? Exploring Machine Learning in Day Trading*, applies the insights gained from the survey to a real-world, trading problem. In it, I implement and evaluate a ML model designed to replicate and surpass human day trading strategies by using second-by-second trade and quote data from all U.S. equities over a two-year period.

1.3 Contributions of This Work

This thesis contributes to ML literature in time series forecasting through both a comprehensive review and practical application. The survey of time series prediction addresses a notable gap in the existing literature. Previous surveys face limitations in comparative analysis because they analyze independent studies which each utilize a different implementation and dataset. This heterogeneity prevents apples to apples comparison. My initial research indicated the superior performance of two classes of ML models: tree-based machine learning (TBML) and deep learning (DL) methods. Consequently, this study only reviews research that compares at least one state-of-the-art TBML and DL methods within the same uniform experimental settings so that nuanced conclusions can be drawn. These conclusions include investigating in key factors that influence model performance, such as the type of time series task, dataset size, time interval of historical data, potential biases in model development, and trade-offs between computational costs and performance.

Key findings of this survey show that specialized TBML architectures like LightGBM and recurrent neural network DL architectures like long short-term memory models consistently outperform other alternatives, with TBML methods showcasing a notable advantage in computational efficiency. Additionally, this study emphasizes that the quality of data is the most influential factor affecting model performance, overshadowing the incremental benefits of hyperparameter tuning. Thus, considerable emphasis is placed on feature selection and engineering in the second paper. Similarly, this study found that combining models and leveraging diverse sources of information further boosts forecasting performance, an insight that heavily informed the construction of the trading model in the second paper.

The second paper applies insights from the survey to develop and evaluate ML models specifically for a day trading application. Day trading refers to the short-term buying and selling of financial instruments within a single trading day. Day trading is a risky endeavor and necessitates rapid decision-making and high-dimensional data processing capabilities. Only the top performing day trading humans actually make a profit, and thus in this paper I create a model that mimics the strategies used by human day traders while leveraging the unique advantages of ML processing with the goal of surpassing the human trading benchmark.

Existing research in this area typically relies on models tailored to individual stocks, limited to longer time horizons, and often lacking in feature diversity. In contrast, human day traders face different constraints as they are prone to emotional decision-making, have slower processing speeds, and can only monitor a few stocks at a time. The ML approach presented in this study overcomes these limitations by simultaneously analyzing the entire universe of U.S. equities and can manage many positions at one time. This study advances the current literature by addressing key gaps in ML applications, leveraging a combination of models trained on second-by-second trade and quote data, and operating on a comprehensive, high-dimensional dataset.

In this paper, I conduct a series of experiments to test various regularization techniques, risk-reward horizon weighting strategies, and execution pricing methods. I evaluate models using both idealized close prices and realistic execution prices derived from the bid and ask prices at which investors are willing to transact, with both methods resulting in profitable models. The model that performed best under realistic trading environments incorporated a min-max regularized target feature based on a risk-reward ratio and applied an equal weighting scheme in this risk-reward calculation. The results of this model show an average profit of 20,000 bps per day with a Sharpe ratio of 15.78 across an average of 999 trades per day. This performance is driven by the model's ability to simultaneously manage a large number of positions at the same time.

The findings from this research show that the ML model I developed achieves performance far exceeding the profitability of top human day traders, demonstrating average daily returns more than 500 times higher. Additionally, the methodological contributions provide a comprehensive, practical blueprint for high-frequency data acquisition, preprocessing, and feature engineering, facilitating replication and extension of the study by future researchers.

1.4 Structure of the Thesis

The remainder of this thesis is organized as follows: Chapter 2 presents a detailed survey on ML methods for time series prediction, identifying best practices and critical factors that influence model effectiveness. Chapter 3 introduces the empirical research of applying these methodologies to the challenge of day trading, detailing data methodologies, model development, and performance evaluation. Finally, Chapter 4 concludes with a summary of the contributions and recommendations for continued research in this area.

CHAPTER 2

A Survey of Machine Learning Methods for Time Series Prediction[1]

[1] Timothy Hall. Submitted to *Applied Sciences, 4/15/25*

Abstract

This study provides a comprehensive survey of the top-performing research papers in the field of time series prediction, offering insights into the most effective machine learning techniques, including tree-based, deep learning, and hybrid methods. It explores key factors influencing model performance, such as the type of time series task, dataset size, and the time interval of historical data. Additionally, the study investigates potential biases in model development and weighs the trade-offs between computational costs and performance. A detailed analysis of the most used error metrics and hyperparameter tuning methods in the reviewed papers is included. Furthermore, the study evaluates the results from prominent forecasting competitions, such as M5 and M6, to enrich the analysis. The findings of this paper highlight that tree-based methods like LightGBM and deep learning methods like recurrent neural networks deliver the best performance in time series forecasting, with tree-based methods offering a significant advantage in terms of computational efficiency. The paper concludes with practical recommendations for approaching time series forecasting tasks, offering valuable insights and actionable strategies that can enhance the accuracy and reliability of predictions derived from time series data.

2.1 Introduction

Time Series Prediction is the process of predicting a future value based on historical sequential observations. Accurate predictions based on time series data plays a crucial role in a wide range of domains where forecasting future values is essential for strategic-planning, resource management, and decision-making. Applications of time series prediction span numerous fields, including electricity consumption forecasting, environmental quality assessments (e.g., air and water quality), meteorological predictions (e.g., rainfall, solar radiation, and wind patterns), medical diagnostics (e.g., forecasting COVID-19 case trends and pneumonia incidences), traffic flow prediction, and financial domains like sales forecasting and stock market analysis.

In recent years, models based on Machine Learning (ML) have demonstrated the most success in time series forecasting and are able to generalize well to unseen data, unlike models based solely on probability and statistics. Specifically, Tree-Based Machine Learning (TBML) and Deep Learning (DL) have emerged as the most prominent approaches, excelling in scenarios where complex, nonlinear dependencies exist within the data. Their ability to generalize to unseen data makes them highly applicable to real-world problems with diverse and dynamic characteristics.

While numerous studies have examined these techniques within specific domains, and several survey papers (Lim & Zohren, 2021; Z. Liu et al., 2021) have analyzed various approaches to time series prediction across domains, existing literature reviews face a significant limitation. Current survey papers struggle to draw meaningful comparisons between models because they analyze independent studies, each utilizing different implementations and datasets. This heterogeneity in experimental setups prevents direct model comparisons and obscures true performance differences. This paper addresses this gap by exclusively reviewing studies that compare both TBML methods and DL approaches within the same experimental framework. By focusing on research where both methodologies are implemented and evaluated by the same researchers using identical datasets, this survey enables more robust conclusions about the relative strengths and weaknesses of these modeling approaches.

The remainder of this paper is organized as follows: **Section 2.2** outlines the methodology employed in this survey. **Section 2.3** reviews TBML architectures, while **Section 2.4** examines DL architectures. **Section 2.5** presents experimental results and discusses findings. **Section 2.6** highlights recent time series prediction competitions, and **Section 2.7** concludes the paper.

2.2 Methodology

A rigorous and systematic methodology was employed to identify the most relevant research papers for this survey. Web of Science was selected as the primary database because it is a trusted publisher-independent source of data with comprehensive coverage of peer-reviewed scientific literature. Given the objective of comparing TBML methods with DL approaches in time series prediction, this paper establishes specific inclusion criteria for article selection:

1. **Focus on Time Series Applications:** The research must address problems involving time series data.
2. **Utilization of Advanced TBML Methods:** Studies must implement advanced TBML architectures, particularly gradient-boosted decision trees or similar structures (e.g., XGBoost, LightGBM, or CatBoost).
3. **Utilization of Advanced Neural Network (NN) Architectures:** Papers must explore sophisticated NN architectures, including but not limited to recurrent neural networks (RNN), feedforward neural networks (FFNN), convolutional neural networks (CNN), long short-term memory networks (LSTM), gated recurrent units (GRU), or Transformers.
4. **Direct Comparisons Using Identical Datasets:** The research must present comparative evaluations of at least one TBML and one DL architecture under identical experimental setups, ensuring consistent datasets and conditions.

To identify relevant literature, a comprehensive search query was developed as follows:

(ALL=(XGBoost) OR ALL=(LightGBM) OR ALL=(CatBoost) OR ALL=("gradient boost*")) AND (ALL=("time series") OR ALL=("time sequence") OR ALL=("temporal series") OR ALL=("temporal sequence") OR ALL=("time forecast*")) AND (ALL=("LSTM") OR ALL=("ANN") OR ALL=("CNN") OR ALL=("RNN") OR ALL=("transformer") OR ALL=("GRU") OR ALL=("deep neur*") OR ALL=("deep lear*"))

This query yielded 589 papers published between 2017 and 2024. To ensure focus on the most influential research, papers were initially selected based on citation count. From the top 100 most-cited papers, 65 articles satisfied the inclusion criteria. To maintain contemporary relevance and capture recent developments in the field, additional temporal criteria were implemented: a minimum of 10 papers per year from 2020 to 2024 must be included. Consequently, 4 highly cited papers from 2023 and the 10 most-cited articles from 2024 were incorporated. In total, this survey encompasses 79 influential papers investigating the comparative performance of TBML and DL approaches in time series analysis (see Appendix A for the complete list of included studies).

2.3 Tree-Based Machine Learning Architectures

This section will present an overview of the best performing TBML Architectures, which are widely utilized for both regression and classification tasks. These include Random Forests (RF), Gradient Boosted Decision Trees (GBDT), and three prominent implementations of GBDT: XGBoost, LightGBM, and CatBoost. Figure 2.1 provides a comparative visualization of the structural differences between RF and GBDT.

2.3.1 Random Forests

Random Forests (RF) is an ensemble learning method that constructs multiple decision trees and combines their outputs through averaging for regression tasks or majority voting for classification tasks. RF uses bootstrapping to train individual decision trees on a random subset of the data at each split. This randomization, coupled with its ensemble nature, enhances the robustness of RF compared to single decision trees, significantly reducing the risk of overfitting. The most widely used library for RF implementation in the studies reviewed in this paper is Scikit-learn.

2.3.2 Gradient Boosted Decision Trees

Gradient Boosted Decision Trees (GBDTs) is a machine learning algorithm that aggregates predictions from multiple weak learners, typically decision trees. GBDTs use “boosting” to build models iteratively where each subsequent models focuses on correcting mistakes made by previous models. The algorithm achieves this by optimizing a differentiable loss function using gradient descent. GBDTs are suitable for both classification tasks, using loss functions such as cross-entropy, and regression tasks, using loss functions like mean squared error. Key hyperparameters to tune in GBDT include tree depth and learning rate, which are crucial for balancing model complexity and reducing overfitting. While Scikit-learn provides a general implementation of GBDT, the three most prominent and high-performing implementations discussed in this survey are XGBoost, LightGBM, and CatBoost, each offering distinct advantages.

2.3.2.1 XGBoost

XGBoost, eXtreme Gradient Boosting (Chen & Guestrin, 2016), introduced by Tianqi Chen in 2014, was designed to address key limitations of traditional GBDT, particularly computational efficiency and scalability. XGBoost gained immediate popularity due to its significant speed improvements, achieved through innovative approaches in decision tree construction. Unlike the greedy splitting methods used in standard GBDT, XGBoost employs a similarity score to evaluate potential splits. This score measures the homogeneity of observations within a node relative to the target variable, assessing the gain provided by a split. To further reduce overfitting, XGBoost incorporates several techniques, including pruning, where branches with a gain below a threshold (hyperparameter gamma, γ) are removed. Similarly, XGBoost has several regularization techniques that prevent overfitting by penalizing complex decision trees. XGBoost also supports advanced features such as built-in cross-validation and highly scalable parallel processing, making it ideal for large-scale datasets.

2.3.2.2 LightGBM

LightGBM (G. Ke et al., 2017), developed by Microsoft in 2017, shares many foundational principles with XGBoost with an even larger focus on computational efficiency. LightGBM achieves superior speed by employing histogram-based binning, which discretizes continuous features into bins, trading minor accuracy losses for dramatic speed gains. Additionally, LightGBM introduces Exclusive Feature Bundling, which is particularly effective for many real-world datasets with high-dimensional sparse features because it consolidates mutually exclusive features into a single representation. Another innovation is Gradient-

Based One-Side Sampling, which prioritizes instances with large gradients while randomly sampling smaller gradients, optimizing learning efficiency. LightGBM also utilizes a leaf-wise tree growth strategy, as opposed to the level-wise growth used in traditional GBDT and XGBoost. This approach selectively grows the leaf with the greatest potential to improve the model, enabling faster convergence and improved accuracy.

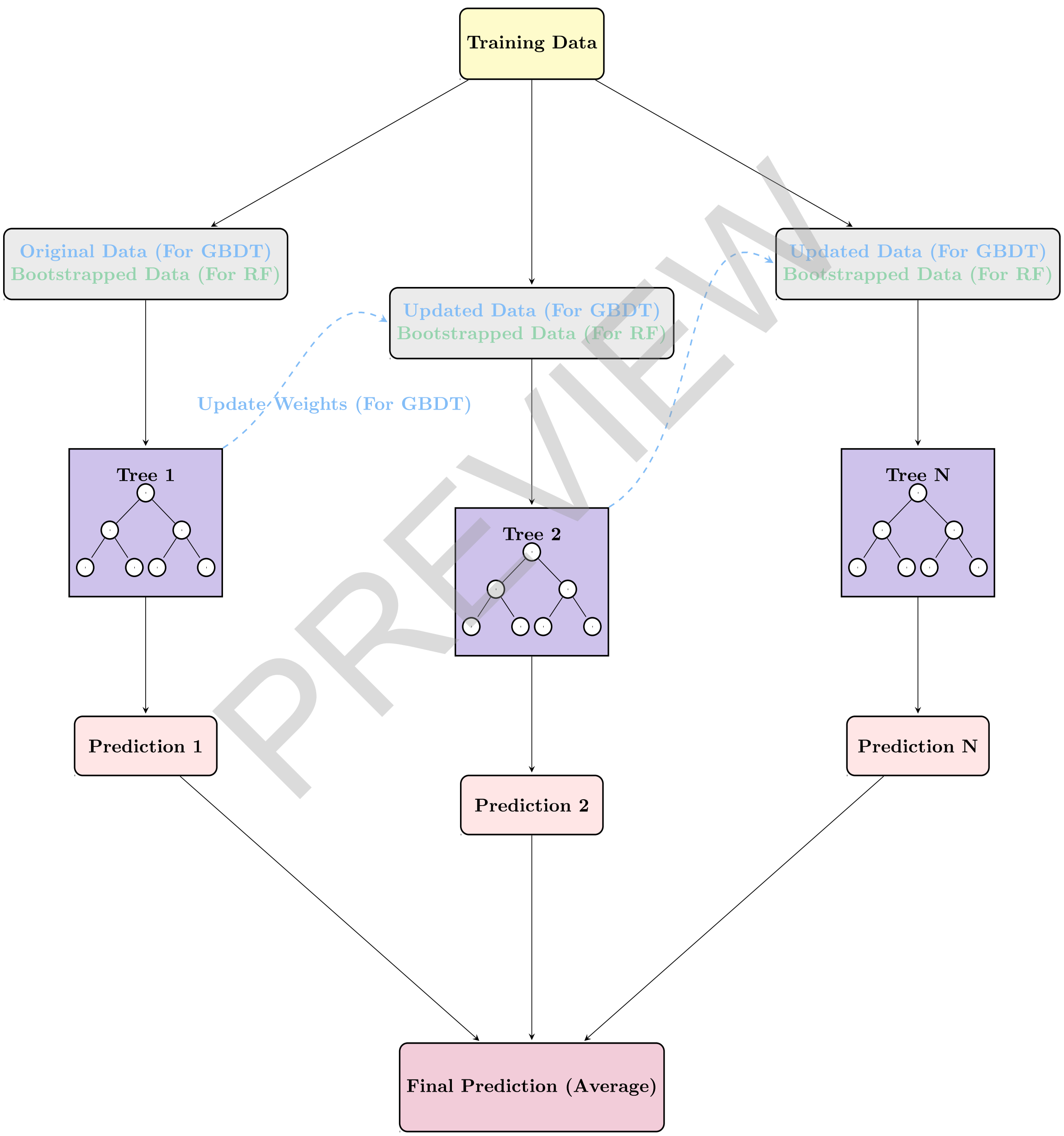

Figure 2.1: Illustrates the architectural differences between RF and GBDT, highlighting attributes unique to RF in green and those specific to GBDT in blue.

2.3.2.3 CatBoost

CatBoost (Prokhorenkova et al., 2018), developed by Yandex in 2017, was designed with a specific emphasis on handling datasets with categorical features. One of its key innovations is a unique implementation of target encoding which takes the concept of traditional target encoding, where categorical values are replaced with the mean of the target variable for each category, and instead constructs the encoding process using only previous data to avoid data leakage. Another distinguishing feature of CatBoost is its use of symmetric decision trees, where all leaves at the same depth use identical splitting criteria. This structure not only accelerates training but also significantly reduces inference time, an essential advantage in some time-sensitive applications of real-world forecasting.

2.4 Deep Learning Architectures

This section presents an overview of the most prominent Deep Learning (DL) architectures encountered in the surveyed literature. DL will be used in this paper to describe a subset of machine learning that utilizes neural networks to perform classification and regression tasks. The architectures are categorized into four primary groups: Feed-Forward Neural Networks (FFNN), Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), and Attention-based architectures. Figure 2.2 presents a visual representation of a FFNN, highlighting the architectural modifications required to transform it into a RNN or CNN.

2.4.1 Feed-Forward Neural Networks

The term Feed-Forward Neural Network (FFNN) is often used interchangeably with other terminologies in the literature, including Artificial Neural Network (ANN), Neural Network (NN), Multilayer Perceptron (MLP), Back Propagation Neural Network (BPNN) (Nan et al., 2022; Pan et al., 2023), Deep Neural Network (DNN), and Deep Feed-Forward Neural Network (DFFNN). These networks are characterized by